

PES UNIVERSITY

Department of Computer Science and Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #10: Student Club Event Ticketing & Budget Portal

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1. Problem Context & Overview

Student clubs require an integrated workflow to submit event proposals, request university budget allocations through approval hierarchies, and generate secure QR-code tickets for attendees.

Target Actors: Club Lead, Campus Admin, Faculty Coordinator, Finance Office and Dean.

2. Functional Requirements

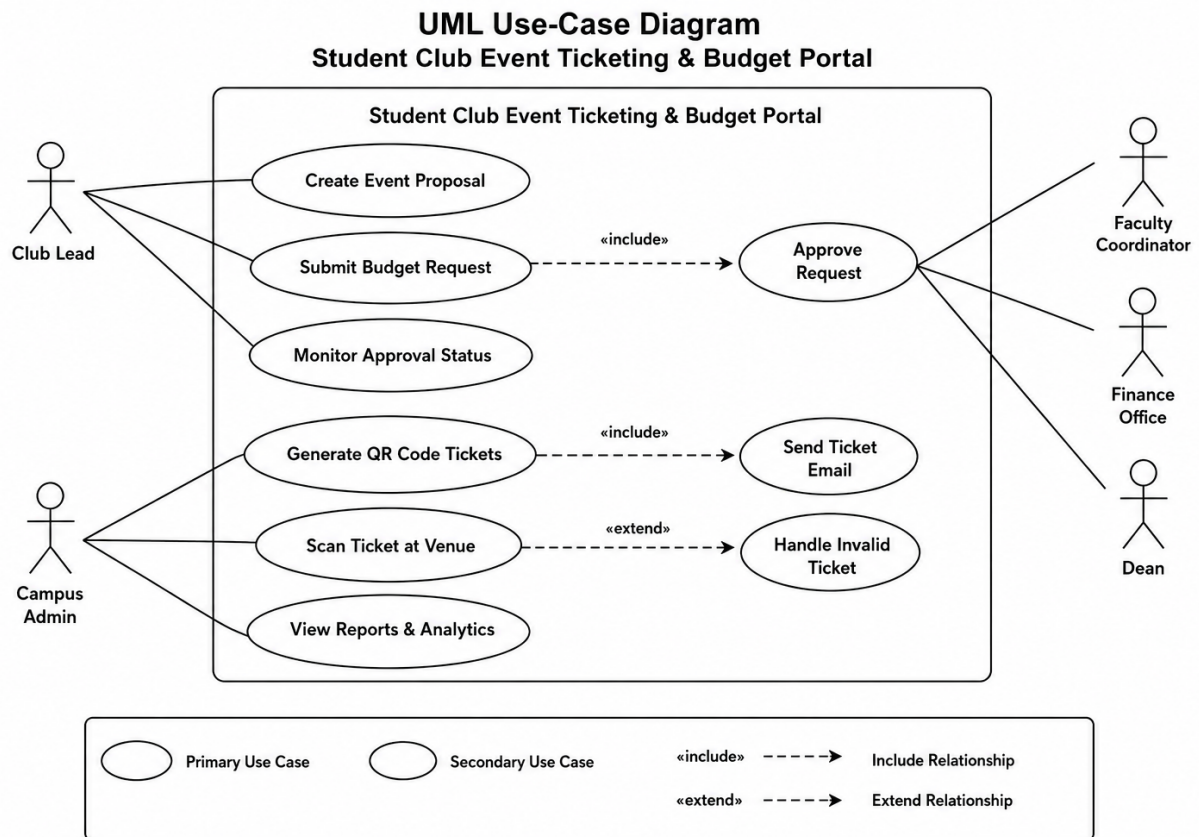
ID	Type	Priority	Description	Acceptance Criteria	Rationale
FR-001	Business Rule	High	The system shall route event budget proposals through a multi-stage approval workflow (Faculty Coordinator to Finance Office to Dean).	Pass: Status transitions sequentially upon digital sign-off. Fail: Request reaches Dean without finance coordinator sign-off.	Ensures proper approval and financial control.
FR-002	Data	High	The system shall allow Club Leads to create and submit event proposals containing event details, date, venue and estimated budget.	Pass: Complete proposal is successfully submitted. Fail: Incomplete mandatory details prevent submission.	Defines the event information that must be captured.
FR-003	Process	High	The system shall generate a unique QR-code ticket for each registered attendee after successful ticket issuance.	Pass: Unique QR ticket is generated. Fail: Duplicate ticket is generated.	Defines the ticket-generation process.
FR-004	Interface	High	The system shall allow Campus Admin to scan QR-code tickets at the venue and validate authenticity and usage status.	Pass: Valid unused ticket grants entry. Fail: Invalid or used ticket is rejected.	Defines the interaction between the administrator and ticket-scanning system.
FR-005	Reporting	Medium	The system shall provide dashboards and reports for events, budgets, approvals and attendee activity.	Pass: Reports accurately reflect system data. Fail: Incorrect or incomplete data is displayed.	Supports monitoring and decision-making.

3. Non-Functional Requirements

ID	Type	Priority	Description	Acceptance Criteria	Rationale
NFR-001	Performance & Security	High	The QR-code ticket scanner API shall validate entry authenticity in under 100 ms at venue check-in points.	Pass: Benchmarking confirms target latency and security standards under simulated peak load. Fail: Response exceeds 100 ms or security checks fail.	Ensures fast and secure check-in.
NFR-002	Usability & Reliability	Medium	The system shall provide 99.5% uptime during event check-in hours and be usable on major mobile browsers.	Pass: Monitoring confirms target availability and core functions work on supported mobile browsers. Fail: Availability or usability target is missed.	Ensures reliable access during events.

4. UML Use-Case Diagram

The diagram below shows the main actors and use cases for the system.



5. Use-Case Flow Specification

Use Case: Submit Budget Request

Primary Actor: Club Lead

Supporting Actors: Faculty Coordinator, Finance Office, Dean

Goal: To submit an event budget request and route it through the approval hierarchy.

Preconditions

- Club Lead is logged into the system.
- Event proposal details have already been created.
- Estimated budget and required supporting details are available.

Postconditions

- Budget request is created and stored.
- Request is sent to the Faculty Coordinator for approval.
- If all stages approve, the request status becomes Approved and the Club Lead is notified.

Main Success Scenario

1. Club Lead selects Submit Budget Request.
2. System displays the budget request form.
3. Club Lead enters the estimated budget and supporting details.
4. Club Lead submits the request.
5. System validates the information and creates the request.
6. System sends the request to the Faculty Coordinator.
7. Faculty Coordinator reviews and approves the request.
8. System sends the request to the Finance Office.
9. Finance Office reviews and approves the request.
10. System sends the request to the Dean.
11. Dean reviews and approves the request.
12. System changes the status to Approved and notifies the Club Lead.

Alternate Flow: Request Rejected

- A1.1 At any approval stage, the approver rejects the request and enters remarks.
- A1.2 System changes the request status to Rejected.
- A1.3 System sends the rejection remarks to the Club Lead.
- A1.4 The use case ends.

6. Relationship Details

«include»: Submit Budget Request includes Approve Request. Generate QR Code Tickets includes Send Ticket Email.

«extend»: Handle Invalid Ticket extends Scan Ticket at Venue when an invalid or already-used ticket is detected.